DS 574 Algorithmic Mechanism Design | **Voting and Social Choice**
Fall 2026 | Prof. Kira Goldner

The Difficulty of Social Choice

We want to choose one outcome from individual rankings, without payments. What should a good rule respect: first choices, the entire ranking, head-to-head majorities, or incentives to report truthfully? These goals lead to different rules, even when everyone reports honestly.

Definition 1. There is a set A of m *alternatives* (candidates) and a set of n voters. Each voter i has a *preference* $\succ_i$, a strict total order on A, and the tuple $\succ = (\succ_1, \ldots, \succ_n)$ is a *preference profile*. A *voting rule* is a function f from profiles to a single winner in A; a *social welfare function* maps every profile to a full ranking of A.

Fix tie-breaking in advance; in the examples below, b wins a tie with a. A rule's output depends on both the ballots and the rule used to combine them.

Plurality: First Choices and Incentives

Plurality. Each voter names one alternative, their first choice; the alternative named most often wins. A majority is not required.

	5 voters	4 voters	2 voters
1st choice	a	c	b
2nd choice	b	d	c
3rd choice	c	b	a
4th choice	d	a	d

The first-choice counts for (a, b, c, d) are $(5, 2, 4, 0)$, so a wins. Five of eleven voters put a first; plurality does not require majority support. *U.S. example.* Maine elects its governor by plurality in the general election, even though it uses RCV in federal general elections.[1]

Can One Voter Benefit by Misreporting?

Definition 2. A rule f is *strategyproof* if no voter ever strictly benefits, according to their true preference, from reporting something other than their true ranking—for every profile, every voter i, and every misreport $\succ_i'$, voter i weakly prefers $f(\succ_i, \succ_{-i})$ to $f(\succ_i', \succ_{-i})$. Otherwise f is *manipulable*.

Historical anchor: Florida, 2000. Bush's certified lead over Gore was 537 votes; Nader received 97,488. This motivates concern about vote splitting, but these single-choice ballots do not identify voters' full rankings or establish who would win under another rule.[2]

[1] Maine Secretary of State, RCV frequently asked questions.

[2] Florida Division of Elections, official 2000 presidential results. The example concerns the state popular-vote contest, not a direct national presidential election.

	4 voters	3 voters	2 voters
1st choice	a	b	c
2nd choice	b	c	b
3rd choice	c	a	a

Truthful counts $(4, 3, 2)$ elect a. One voter with true ranking $c \succ b \succ a$ reports $b \succ c \succ a$. Counts become $(4, 4, 1)$; the fixed tie-break elects b, which that voter prefers to a. Other reports stay fixed: this is a profitable unilateral deviation.

With only two alternatives, majority rule is strategyproof: changing a truthful vote can only move the count toward the less-preferred outcome. The difficulty starts when there are more alternatives.

Borda: Use the Whole Ranking

Borda count. With m alternatives, each ballot awards $m - 1$ points to its first choice, $m - 2$ to its second, down to 0 for last place. The highest total wins. Return to the same eleven voters:

	5 voters	4 voters	2 voters
1st choice	a	c	b
2nd choice	b	d	c
3rd choice	c	b	a
4th choice	d	a	d

The supplied Borda totals are $(17, 20, 21, 8)$ for (a, b, c, d): c wins. The same preferences elect a under plurality and c under Borda. Borda counts positions throughout each ballot; plurality counts only first choices. The two rules aggregate different information, rather than disagreeing over arithmetic. *American ranked-points example.* Heisman Trophy ballots award three, two, and one points to a voter's top three players. This is a Borda-style scoring rule, not the complete-ranking Borda rule analyzed here.[3]

Condorcet: Head-to-Head Majorities

A *Condorcet winner* beats every other alternative in a head-to-head majority comparison. The Condorcet criterion asks a voting rule to select that alternative whenever one exists. Return to the same eleven voters:

	5 voters	4 voters	2 voters
1st choice	a	c	b
2nd choice	b	d	c
3rd choice	c	b	a
4th choice	d	a	d

Candidate b beats a by 6–5 and beats c and d by 7–4. Thus b is the Condorcet winner. The same profile has three different winners: plurality a, Borda c, and head-to-head majority b. Neither plurality nor Borda always satisfies the Condorcet criterion.

[3] Heisman Trophy, "Heisman Balloting: How it works."

What If Majority Comparisons Cycle?

	Voter 1	Voter 2	Voter 3
1st choice	a	b	c
2nd choice	b	c	a
3rd choice	c	a	b

Each contest is 2–1: a beats b, b beats c, and c beats a. Every voter's ranking is transitive, but the majority relation is not. No candidate beats everyone, so a rule that always returns a winner needs a further decision in this case.

Instant Runoff: Elimination and Transfers

Instant-runoff voting (IRV), called RCV here. If a candidate has a strict majority of the active first choices, elect them. Otherwise eliminate the candidate with the fewest first choices and transfer each of their ballots to its highest-ranked remaining candidate. Repeat, with tie-breaking fixed beforehand. Our example uses complete rankings. *U.S. examples.* Maine uses RCV for federal general elections and state primaries, but not general elections for governor or the state legislature. Alaska pairs a top-four primary with RCV general elections (first used in 2022). NYC uses RCV for municipal primaries and special elections, first used in 2021, rather than its general elections.[4] *A different route to a majority.* In Georgia U.S. Senate elections, there is a separate top-two runoff if no candidate wins the required majority. This is a second election, not instant elimination and transfers from one ranked ballot.[5]

	6 voters	5 voters	4 voters	2 voters
1st choice	a	c	b	b
2nd choice	b	a	c	a
3rd choice	c	b	a	c

In order (a, b, c), the first counts are $(6, 6, 5)$. Eliminate c; its five ballots transfer to a. Candidate a wins 11–6.

Now raise a above b on the two-voter bloc's ballots, changing no other comparison. The first counts become $(8, 4, 5)$. Eliminate b; its four ballots transfer to c. Candidate c wins 9–8. More first-choice support for a changed who was eliminated and made a lose.

A rule is *monotone* if raising its winner on ballots, with all other comparisons unchanged, cannot make that candidate lose. This example shows that IRV is not monotone. It is a concrete elimination-order difficulty, not itself a proof that a voter benefits by lying.

A General Obstacle

Theorem 1 (Gibbard–Satterthwaite; stated without proof [**??**])**.** *With at least three alternatives, every deterministic, onto, strategyproof voting rule on all strict preference profiles is a dictatorship.*

[4] Maine Secretary of State; Alaska Division of Elections; NYC Votes; NYC Campaign Finance Board, 2021–2022 report.

[5] State of Georgia, "Vote in Runoff Elections."

Here *onto* means every alternative wins at some profile; a dictatorship always chooses one fixed voter's top alternative. The theorem assumes all strict preference profiles are allowed. It concerns a single winner and profitable misreports, not every possible fairness property. It explains why changing the counting rule alone cannot meet all these requirements.

Single-Peaked Preferences and the Median Rule

Suppose at least three alternatives in a finite set A lie on one publicly fixed line, such as possible levels of spending on a single public service. We restrict preferences rather than changing the alternatives or introducing payments.

Definition 3. A strict ranking is *single-peaked* on this line if it has a favorite alternative p_i, called voter i's *peak*, and alternatives become less preferred as we move away from p_i on either side. Formally, $x < y \leq p_i$ implies $y \succ_i x$, and $p_i \leq x < y$ implies $x \succ_i y$. The condition does not determine comparisons between opposite sides of the peak; preferences need not be based on distance.

Let $n = 2k + 1 \geq 3$ be odd. Each voter reports a peak in the fixed ordered set A. The *median rule* chooses the middle reported peak, counting repetitions.

Theorem 2. *On the domain of strict single-peaked preferences on a common fixed order, reporting the true peak is a dominant strategy for the median rule.*

The useful comparison holds the other reports fixed and asks which outcomes one voter can cause, rather than checking many possible lies.

Proof. Fix voter i and sort the other $2k$ reported peaks as $q_1 \leq \cdots \leq q_{2k}$. Write $L = q_k$ and $U = q_{k+1}$ for the middle two. If i reports r, the median is

$$f(r, q_{-i}) = \begin{cases} L & r \leq L, \\ r & L \leq r \leq U, \\ U & r \geq U. \end{cases}$$

Thus the possible outcomes are precisely the alternatives of A between L and U. If the true peak p_i lies in this interval, truth gives p_i, their favorite alternative. If $p_i < L$, every possible outcome is on the right of the peak, so single-peakedness makes L best; truth gives L. If $p_i > U$, every possible outcome is on the left, so U is best; truth gives U. No report yields a strictly better outcome. □

For example, when the other four peaks are $2, 4, 6, 8$, the voter can select an outcome only between 4 and 6. A true peak of 5 gets 5; a true peak below 4 gets 4; a true peak above 6 gets 6. The same interval argument, not a list of deviations, proves all three cases.

There is no contradiction with Gibbard–Satterthwaite: the median rule is strategyproof on the restricted single-peaked domain, not on all strict preference profiles. It is onto (all voters can report the same alternative) and no fixed voter dictates the outcome. The restriction to a common order is doing the work.[6]

[6] See also Tim Roughgarden, CS269I Lecture 3, for the single-peaked median rule.

What the Examples Establish

Plurality can reward abandoning a favorite. Positional scores can disagree with head-to-head majorities. Majority comparisons can cycle. Elimination can respond perversely to increased support. Gibbard–Satterthwaite gives a general incentive obstacle, while the median rule gives a positive construction on a restricted domain. The claim is not that every rule fails in the same way, or that preferences are always single-peaked.

Reference: IIA and the Borda Guarantee

The following formal properties supplement the examples above. A *Condorcet loser* loses every head-to-head majority contest against another alternative.

Definition 4. A social welfare function satisfies *independence of irrelevant alternatives* (IIA) if, for every pair x, y, the relative order of x and y in the social ranking depends only on each voter's relative order of x and y—never on where any third alternative z sits in anyone's ranking.

Exercise. The Borda scores (a, b, c) are initially $(10, 8, 3)$. Moving c to last place on the three-voter bloc's ballots gives $(10, 11, 0)$. What stays fixed about a and b? Explain why their reversed social order violates IIA.

	4 voters	3 voters
1st choice	b	a
2nd choice	a	c
3rd choice	c	b

Initially the scores are $a : 10$, $b : 8$, $c : 3$. After the change, the three-voter bloc reports $a \succ b \succ c$, giving scores $a : 10$, $b : 11$, $c : 0$. Every voter's order of a and b is unchanged, but Borda reverses their social order. Hence Borda violates IIA on a fixed set of alternatives.
Let $B(x)$ denote alternative x's total Borda score. Each ballot gives x one point for each alternative ranked below x.

Proposition 3. *The Borda winner is never the Condorcet loser.*

Proof. Observe first that an alternative's Borda score can be written as $B(x) = \sum_{y \neq x} \#\{i : x \succ_i y\}$, since x earns one point for each opponent it is ranked above on each ballot. Every pairwise contest contributes exactly n points, split between its two participants, so

$$\sum_{x \in A} B(x) = n\binom{m}{2} = \frac{n\,m(m-1)}{2}, \qquad \text{average} = \underbrace{\frac{n(m-1)}{2}}_{n/2 \text{ per opponent}}.$$

Now suppose x is the Condorcet loser. Then $\#\{i : x \succ_i y\} < n/2$ for each of the $m - 1$ opponents y, so $B(x) < \frac{n(m-1)}{2}$: the Condorcet loser scores strictly below average. Since some alternative must score at least the average, x is not the Borda winner. □

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